

Stability and synchronization of discrete-time fractional-order reaction–diffusion neural networks

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ABSTRACT

Discrete reaction–diffusion models have garnered considerable interest for their effectiveness in numerical simulations and their capacity to accurately reflect complex physical phenomena. Unlike traditional models, they offer a refined depiction of system dynamics across various contexts. This paper presents a pioneering investigation into the stability and synchronization of discrete-time fractional-order neural networks incorporating reaction–diffusion terms-atopic that has not been previously addressed in the literature. Leveraging Lyapunov stability theory, we derive novel global uniform asymptotic stability conditions. Furthermore, we propose new synchronization criteria based on linear state feedback control, providing a systematic framework for ensuring coordinated behavior in such networks. The effectiveness and applicability of the proposed theoretical results are validated through comprehensive numerical simulations, highlighting the practical relevance of this study.

1. Introduction

Artificial neural networks (NNs) have gained widespread attention due to their extensive applications in various fields, including medical imaging, face recognition, secure communications, and pattern recognition [1–3]. Neural networks mimic the structure and function of the human brain, characterized by their ability to process large amounts of data and complex patterns in parallel. Their unique learning capabilities and nonlinear mapping abilities make them powerful tools for solving complex problems. As two fundamental issues in control theory [4], many excellent research results on the stability and synchronization of neural networks have been reported [5–8].

In recent years, fractional calculus has been increasingly incorporated into neural network theory because of its capability to capture nonlocal memory and hereditary effects observed in biological and cognitive systems. As a result, the study of fractional neural networks (FNNs) has grown rapidly, with researchers exploring their dynamical behaviors [9–12]. Synchronization has become a key research focus due to its relevance to coordinated information processing, pattern formation, and neuromorphic intelligence applications, leading to multiple meaningful contributions in the field [13,14]. Furthermore, recent work has demonstrated that fractional models are not only theoretically valuable but also practically applicable fractional epidemic modeling has shown improved accuracy in characterizing memory-driven COVID-19 transmission dynamics [15], while numerical schemes such as the L1 finite-difference method on graded meshes ensure stable and accurate solutions for multi-term fractional systems [16]. These

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developments collectively indicate that research on fractional-order synchronization is supported by real computational feasibility and demonstrated real-world applicability.

In addition to the above developments, a series of recent studies have further advanced fractional-order system theory in the contexts of complex network dynamics, robustness analysis, and generalized control frameworks. For example, several works have proposed observer-based output feedback strategies for uncertain fractional-order systems, while others have investigated the dynamic behaviors of state-dependent fractional-order systems under multiple external disturbances [17]. Methods based on the Mittag-Leffler function have also been introduced for analyzing fractional-order complex networks [18]. Furthermore, fractional-order models have been extended to a variety of nonlinear settings, including discrete-time systems, fuzzy systems [19], and memristive systems [20]. These results collectively demonstrate the rapid expansion of fractional-order dynamical methodologies across broader application domains [21,22]. However, it is important to note that although these studies enrich the theoretical landscape of fractional-order systems, none of them address reaction–diffusion neural network models that simultaneously incorporate fractional-order memory effects and spatial diffusion within a discrete-time framework. This reveals a clear gap in the current frontier of fractional-order research.

Many models in the real world must consider spatial diffusion factors, which is particularly important in areas such as food distribution, virus transmission, biological models, and network models. Especially in engineering applications, neural networks are often implemented through electronic circuits. Due to stochastic behavior and interactions with the surrounding medium, the movement of electrons can often be characterized by diffusion processes, particularly in thermally fluctuating or heterogeneous environments. Therefore, it becomes particularly important to study neural networks containing spatial diffusion terms. This topic has attracted considerable attention in recent years due to its ability to more accurately describe and predict the behavior of complex systems [23–26]. For example, in biological models, cell interactions depend on spatial diffusion; in virus transmission, it helps simulate disease spread more accurately; and in engineering, analyzing electron behavior in electromagnetic fields aids circuit optimization. By introducing the spatial diffusion term, the researchers not only enriched the theoretical model, but also provided more effective analysis tools and methods, opening up new directions for the research and application of FNN [27–29].

However, most existing studies on fractional-order reaction–diffusion neural networks have focused on continuous-time formulations. These approaches, while theoretically rich, are difficult to implement in digital or sampled-data systems. Moreover, the differentiation of Lyapunov functions in continuous fractional-order models poses analytical challenges when extended to spatiotemporal dynamics. Therefore, developing a discrete-time fractional-order reaction–diffusion model is crucial for bridging theoretical analysis with practical computation and control implementation.

In engineering applications, continuous-time models are often discretized to more effectively evaluate structural behavior, since signals in digital systems are inherently discrete. As a result, discrete-time neural networks have attracted increasing attention [30–32]. With advances in computational technologies, discrete models have shown superior efficiency and applicability compared with continuous models, further promoting research on discrete dynamic neural networks [33–35]. However, most existing studies focus on time-discrete neural networks, while much less attention has been paid to spatially discrete or reaction–diffusion-type fractional-order systems. Although recent works have examined Mittag-Leffler stability, fixed-point approaches, nabla fractional systems, and synchronization of fractional-order discrete networks with delays [12,30–32], these results still show clear limitations. Current methods predominantly rely on classical Lyapunov theory or fixed-point techniques, which are difficult to apply to spatiotemporally coupled or reaction–diffusion discrete fractional networks. In particular, the differentiation of Lyapunov functions in fractional discrete frameworks remains a major technical obstacle.

Although some studies have investigated discrete-time integer-order neural networks or continuous-time fractional-order systems, there remains a significant gap in understanding discrete-time fractional-order reaction–diffusion neural networks (FDRDNNs). Specifically, previous works did not simultaneously account for both fractional dynamics and spatial diffusion under discrete-time formulations. Hence, our study fills this gap by proposing a unified framework that jointly considers these two effects. Based on the above analysis, this paper established a discrete-time fractional-order reaction–diffusion model. The main contributions of this study are as follows:

1. This study develops the first unified model integrating fractional-order memory, spatial diffusion, and discrete-time dynamics, bridging the theoretical gap between continuous fractional systems and discrete neural network research.
2. By leveraging the properties of discrete fractional difference operators, we establish a new integral inequality that serves as a fundamental analytical tool for stability and synchronization in discrete reaction–diffusion processes.
3. Within this framework, we rigorously derive global uniformly asymptotic stability conditions for discrete FDRDNNs, effectively extending classical Lyapunov theory to the discrete fractional domain.
4. We further introduce synchronization criteria based on linear state-feedback control, offering robust and scalable strategies for stabilizing spatiotemporally coupled networks in neuromorphic, biological, and engineering applications.

Overall, this work integrates three essential perspectives fractional dynamics, spatial diffusion, and discrete-time realization into a cohesive framework for analyzing the stability and synchronization of neural networks. This integration not only advances the theoretical understanding of FDRDNNs but also provides practical insights for neuromorphic computation and biological modeling.

The remainder of this paper is organized as follows: Section 2 presents several definitions, relevant lemmas, and the establishment of the model. In Section 3, stability conditions are derived, and synchronization criteria under the controller are obtained. Section 4 provides numerical simulations to verify the theoretical results. Finally, Section 5 offers conclusions and prospects for future research.

2. Preliminaries and model description

Notations: We define $(h\mathbb{N})_0 = \{j \mid j = 0, h, 2h, \dots\}$. Let $\mathbb{R}^n$ be the set of all n -dimensional real vectors. Define

$$\Omega = \{hm = (hm_1, hm_2, \dots, hm_n)^T \in \mathbb{R}^n \mid m \in \mathbb{Z}^n, |hm_i| \leq b_i, i = 1, 2, \dots, n\},$$

which is an open bounded domain with smooth boundary $\partial\Omega$. For any function $v : \Omega \times (h\mathbb{N})_0 \rightarrow \mathbb{R}^n$, define

$$\Delta_h^2 v(hm, \cdot) = \frac{v(hm + h, \cdot) - 2v(hm, \cdot) + v(hm - h, \cdot)}{h^2},$$

and the backward second-order finite difference operator

$$\nabla_h^2 v(hm, \cdot) = \frac{v(hm, \cdot) - 2v(hm - h, \cdot) + v(hm - 2h, \cdot)}{h^2}.$$

Then it follows that

$$\Delta_h^2 v(hm, \cdot) = \nabla_h^2 v(hm + h, \cdot).$$

The operator Δ_h^2 is called the second-order central finite difference with respect to the space variable. Sometimes, $v = (v_1, v_2, \dots, v_n)^T$ is used for simplicity.

In this paper, the Caputo-type discrete fractional h -sum and nabla fractional difference operators are defined following the standard framework in discrete fractional calculus [36]. These definitions ensure the nonlocal property and reduce to the classical difference operators when $\theta \rightarrow 1$.

Definition 1 ([36]). Suppose $x(t) : (h\mathbb{N})_0 \rightarrow \mathbb{R}$, the Caputo fractional h -sum with point 0 is defined as

$${}_0^C \nabla_h^{-\theta} x(t) = \sum_{v=1}^{\frac{t}{h}} \frac{(t - \rho(vh))_h^{\overline{\theta-1}}}{\Gamma(\theta)} x(vh)h,$$

where $t \in (h\mathbb{N})_h$, $0 < \theta < 1$, $t_h^{\overline{\theta}} = h^{\overline{\theta}} \frac{\Gamma(\frac{t}{h} + \theta)}{\Gamma(\frac{t}{h})}$ and $\rho(vh) = vh - h$.

Definition 2 ([36]). For any function $x(t) : (h\mathbb{N})_0 \rightarrow \mathbb{R}$, $t \in (h\mathbb{N})_h$, the Caputo nabla fractional difference operator of $x(t)$ of order h is defined as follows

$${}_0^C \nabla_h^{\theta, \mu} x(t) = \sum_{v=1}^{\frac{t}{h}} \frac{(t - \rho(vh))_h^{\overline{-\theta}}}{\Gamma(1 - \theta)} \nabla_{vh} x(vh)h,$$

in which $\nabla_{vh} x(vh) = \frac{x(vh) - x(vh-h)}{h}$.

Definition 3. For any $x(t, hm) : (h\mathbb{N})_0 \times \Omega \rightarrow \mathbb{R}$, $t \in (h\mathbb{N})_h$. We define θ -order Caputo nabla fractional difference of $x(t, hm)$ as follows

$${}_0^C \nabla_h^{\theta} x(t, mh) = \sum_{v=1}^{\frac{t}{h}} \frac{(t - \rho(vh))_h^{\overline{-\theta}}}{\Gamma(1 - \theta)} \nabla_{vh} x(vh, mh)h,$$

in which $t \in (h\mathbb{N})_h$ and $\nabla_{vh} x(vh, mh) = \frac{v(vh, mh) - v(vh-h, mh)}{h}$.

Lemma 1 ([30]). Let $x(t) = (x_1(t), x_2(t), \dots, x_n(t))^T$ be function, then the following difference inequality holds

$${}_0^C \nabla_0^{\theta} x^2(t) \leq 2x(t) {}_0^C \nabla_0^{\theta} x(t), \quad t \in \mathbb{N}_h.$$

Lemma 2. Let

$$\Omega = \{hm = (hm_1, \dots, hm_n)^T \in \mathbb{R}^n : |hm_i| \leq b_i, i = 1, \dots, n\},$$

and let $v(hm, \cdot)$ be a real-valued function on Ω satisfying either the Dirichlet condition $v|_{\partial\Omega} = 0$ or the Neumann condition $\nabla_{hm_i} v|_{\partial\Omega} = 0$. For the Neumann case, define

$$\bar{v}_i(hm_i) = \frac{1}{2b_i} \int_{-b_i}^{b_i} v(hm_1, \dots, s, \dots, hm_n) \nabla_{hm_i} s, \quad w := v - \bar{v}_i.$$

Then, for each $i = 1, \dots, n$, the following discrete Poincaré-Wirtinger inequality holds:

$$\int_{\Omega} u^2 \nabla_{hm} \leq b_i^2 \int_{\Omega} (\nabla_{hm_i} v)^2 \nabla_{hm}, \quad u = \begin{cases} v, & \text{Dirichlet case,} \\ v - \bar{v}_i, & \text{Neumann case.} \end{cases}$$

In particular, if $\bar{v}_i \equiv 0$ in the Neumann case, the inequality coincides with the Dirichlet form.

Proof. Step 1 (Dirichlet case) Fix all coordinates except $\hbar m_i$. From the fundamental theorem of difference calculus and the boundary condition $v(\pm b_i) = 0$, we have

$$v(\hbar m) = \int_{b_i}^{\hbar m_i} \nabla_{\hbar m_i} v(\hbar m_1, \dots, \hbar m_n) \nabla_{\hbar m_i}, \quad (1)$$

and

$$v(\hbar m) = - \int_{\hbar m_i}^{-b_i} \nabla_{\hbar m_i} v(\hbar m_1, \dots, \hbar m_n) \nabla_{\hbar m_i}. \quad (2)$$

Adding (1) and (2) and taking absolute values yields

$$2|v(\hbar m)| \leq \int_{-b_i}^{b_i} \left| \nabla_{\hbar m_i} v(\hbar m_1, \dots, \hbar m_n) \right| \nabla_{\hbar m_i}. \quad (3)$$

Applying the Cauchy–Schwarz inequality gives

$$\begin{aligned} 4|v(\hbar m)|^2 &\leq \left(\int_{-b_i}^{b_i} \left| \nabla_{\hbar m_i} v \right| \nabla_{\hbar m_i} \right)^2 \\ &\leq \left(\int_{-b_i}^{b_i} 1 \nabla_{\hbar m_i} \right) \left(\int_{-b_i}^{b_i} (\nabla_{\hbar m_i} v)^2 \nabla_{\hbar m_i} \right) \\ &= 2b_i \int_{-b_i}^{b_i} (\nabla_{\hbar m_i} v)^2 \nabla_{\hbar m_i}. \end{aligned} \quad (4)$$

Integrating both sides of (4) with respect to $\hbar m_i$ over $[-b_i, b_i]$ and then over $\hbar m_i$, we obtain

$$\int_{\Omega} v^2 \nabla_{\hbar m} \leq b_i^2 \int_{\Omega} (\nabla_{\hbar m_i} v)^2 \nabla_{\hbar m}.$$

This completes the proof for the Dirichlet case.

Step 2 (Neumann case). Let $w := v - \bar{v}_i$. For fixed $\hbar m_i$, one has $\int_{-b_i}^{b_i} w \nabla_{\hbar m_i} = 0$. By the fundamental theorem of difference calculus,

$$w(\hbar m) = \int_{-b_i}^{\hbar m_i} \nabla_{\hbar m_i} v(\cdot) \nabla_{\hbar m_i} - \frac{\hbar m_i + b_i}{2b_i} \int_{-b_i}^{b_i} \nabla_{\hbar m_i} v(\cdot) \nabla_{\hbar m_i}. \quad (1')$$

Using the triangle inequality and the fact that $|\hbar m_i + b_i|/(2b_i) \leq 1$, we obtain

$$2|w(\hbar m)| \leq \int_{-b_i}^{b_i} \left| \nabla_{\hbar m_i} v(\hbar m_1, \dots, \hbar m_n) \right| \nabla_{\hbar m_i}. \quad (3')$$

Applying the Cauchy–Schwarz inequality to (3') gives

$$4|w(\hbar m)|^2 \leq 2b_i \int_{-b_i}^{b_i} (\nabla_{\hbar m_i} v)^2 \nabla_{\hbar m_i}. \quad (4')$$

Integrating both sides of (4') over $\hbar m_i$ and $\hbar m_i$ yields

$$\int_{\Omega} w^2 \nabla_{\hbar m} \leq b_i^2 \int_{\Omega} (\nabla_{\hbar m_i} v)^2 \nabla_{\hbar m}.$$

Hence, inequality holds for the Neumann–Wirtinger case. If v additionally satisfies the zero-mean condition $\bar{v}_i \equiv 0$, then $w = v$ and the inequality coincides with the Dirichlet form.

Remark 1. Lemma 2 remains valid for both the Dirichlet and homogeneous Neumann boundary conditions. In the Dirichlet case, $v(\hbar m)|_{\partial\Omega} = 0$ ensures the function itself vanishes on the boundary, while in the Neumann case, $\nabla_{\hbar m_i} v(\hbar m)|_{\partial\Omega} = 0$ implies no flux across the spatial boundary. Under either condition, all boundary terms vanish when performing the discrete integration by parts, and the constant b_i^2 in the inequality remains unchanged. Therefore, Lemma 2 provides a unified discrete Poincaré-type inequality applicable to both boundary conditions, which guarantees the same diffusion-dominated dissipation term in the subsequent stability analysis.

Remark 2. The above integral inequality serves as the discrete analogue of the classical Poincaré inequality and plays a crucial role in the analysis of discrete-time fractional-order systems. Specifically, it provides a mathematical bridge between the spatial diffusion operator and the nonlocal Caputo-type fractional difference operator in the temporal dimension. In the Lyapunov function framework, this inequality enables the transformation of the diffusion-related gradient term $\int_{\Omega} \|\nabla_{\hbar m} v(\hbar m, \cdot)\|^2 \nabla_{\hbar m}$ into a coercive bound on the state energy $\int_{\Omega} v^2(\hbar m, \cdot) \nabla_{\hbar m}$. Consequently, it supplies a negative-definite dissipation term in ${}^C_0 \nabla_h^\theta V(t)$, thereby ensuring that the Lyapunov-based stability and synchronization analysis of discrete fractional neural networks remains rigorous and self-contained.

Lemma 3 ([32]). Let $V(t)$ be non-negative functions and $0 < \theta < 1$, $t \in \mathbb{N}_h$ such that

$${}_0^C \nabla_h^\theta V(t) \leq -\varpi V(t),$$

where $0 < \varpi < h^{-\theta}$, then

$$V(t) \leq V(0)E_h^\theta(-\varpi, t),$$

in which $E_h^\theta(-\varpi, t) = \sum_{i=0}^\infty (-\varpi)^i \frac{t^{\overline{i\theta}}}{\Gamma(i\theta+1)}$.

This article considers the discrete-time fractional neural networks with discrete spatial diffusions, expressed in the following formula

$$\begin{cases} {}_0^C \nabla_h^\theta v_p(hm, t) &= \sum_{j=1}^n d_{pj} \nabla_{hm}^2 v(hm + h, t) - \alpha_p v_p(hm, t) \\ &\quad + \sum_{k=1}^n \beta_{pk} f_k(v_k(hm, t)) + J_p, \\ \nabla_{hm} v_p(hm + h, t) |_{hm \in \partial\Omega} &= 0, v_p(hm, 0) = \varphi_p(hm), (hm, t) \in (\Omega \times (h\mathbb{N})_h). \end{cases} \quad (5)$$

where v_p represents the states of the p -th neuron; and α_p signifies the strength of the p th neuron; d_{pj} represents the diffusion coefficient; β_{pk} denotes the connecting intensities between neurons; $f_k(v_k(hm, t))$ is the activation function; J_p stands for the external inputs; the Neumann boundary condition $\nabla_{hm} v_p(hm + h, t) |_{hm \in \partial\Omega} = 0$ implies no flux across the spatial boundary, which guarantees energy conservation in the diffusion process. The bounded domain Ω and smooth boundary ensure the applicability of Lemma 2.

Remark 3. Wu et al. [12] were the first to introduce discrete fractional-order operators into fractional-order neural networks and systematically analyzed their stability, laying the foundation for modeling and theoretical research in this field. The modeling approach in this paper primarily builds upon their work, extending the framework to accommodate a broader range of applications. Unlike existing studies, this paper introduces discrete-time diffusivity into discrete-time fractional-order neural networks for the first time, capturing the spatial effects of information propagation within the system. This extension not only enriches the theoretical research on fractional-order neural networks but also provides new insights for their applications in neural dynamics, image processing, and complex system modeling.

Remark 4. System (5) represents a more general neural network system, which includes some previous network models as special cases. For instance, when $d_{pi} = 0$, the system degenerates into the general discrete-time fractional-order neural network studied in [12,32].

To qualitatively analyze system (5), we make the following assumptions:

Assumption 1. Assume that the active function $f_k(\cdot)$ meets as follows, for any real numbers m and n , there exists nonnegative number ℓ_k , such that

$$|f_k(m) - f_k(n)| \leq \ell_k |m - n|.$$

Definition 4. System (5) is said to be global uniformly stable, if for any two solutions of system (5) with initial functions φ_p, ω_p , one has

$$\lim_{t \rightarrow \infty} \|v_p(hm, t) - v_p^*(hm, t)\| = 0.$$

3. Main results

In this section, we will investigate the stability and synchronization of discrete-time fractional order neural networks with discrete reaction. diffusion

Theorem 1. Under Assumption 1, define

$$\varsigma := \min_{1 \leq p \leq n} \left\{ 2\alpha_p + \sum_{j=1}^n \frac{2d_{pj}}{b_j^2} - \sum_{k=1}^n (|\beta_{pk}| \ell_k + |\beta_{kp}| \ell_p) \right\}.$$

If $0 \leq \varsigma \leq h^\theta$, then system (5) is global uniformly stable. Moreover, if $\varsigma > 0$, the origin is global uniformly asymptotically stable.

Proof. Let $v_p(hm, t)$ and $v_p^*(hm, t)$ be two solutions of system (5) with initial data $v_p(hm, 0) = \varphi_p(hm)$ and $v_p^*(hm, 0) = \omega_p(hm)$, respectively. Set the error $r_p(hm, t) = v_p(hm, t) - v_p^*(hm, t)$. Then

$${}_0^C \nabla_h^\theta r_p(hm, t) = \sum_{j=1}^n d_{pj} \nabla_{hm}^2 r_p(hm + h, t) - \alpha_p r_p(hm, t) + \sum_{k=1}^n \beta_{pk} f_k(r_k(hm, t)). \quad (6)$$

Consider the Lyapunov functional

$$V(r(\hbar m, t), t) := \sum_{p=1}^n \int_{\Omega} \frac{1}{2} r_p^2(\hbar m, t) \nabla_{\hbar m}.$$

By the fractional product inequality (Lemma 1),

$${}_0^C \nabla_{\hbar}^{\theta} V \leq \sum_{p=1}^n \int_{\Omega} r_p(\hbar m, t) {}_0^C \nabla_{\hbar}^{\theta} r_p(\hbar m, t) \nabla_{\hbar m}.$$

Substituting (6) and splitting terms yields

$$\begin{aligned} {}_0^C \nabla_{\hbar}^{\theta} V &= \sum_{p=1}^n \int_{\Omega} r_p \left(\sum_{j=1}^n d_{pj} \nabla_{\hbar m}^2 r_p(\hbar m + \hbar, t) \right) \nabla_{\hbar m} - \sum_{p=1}^n \int_{\Omega} \alpha_p r_p^2 \nabla_{\hbar m} \\ &\quad + \sum_{p=1}^n \int_{\Omega} \sum_{k=1}^n \beta_{pk} f_k(r_k) r_p \nabla_{\hbar m}. \end{aligned} \quad (7)$$

Diffusion term. Applying the discrete integration-by-parts (distribution summation formula) in each coordinate j and using either homogeneous Neumann boundary conditions (Lemma 2) to cancel boundary fluxes,

$$\int_{\Omega} r_p \nabla_{\hbar m}^2 r_p(\hbar m + \hbar, t) \nabla_{\hbar m} = - \int_{\Omega} (\nabla_{\hbar m} r_p)^2 \nabla_{\hbar m}.$$

Then the discrete Poincaré inequality gives

$$\int_{\Omega} (\nabla_{\hbar m} r_p)^2 \nabla_{\hbar m} \geq \sum_{j=1}^n \frac{1}{b_j^2} \int_{\Omega} r_p^2 \nabla_{\hbar m},$$

hence

$$\sum_{p=1}^n \int_{\Omega} r_p \left(\sum_{j=1}^n d_{pj} \nabla_{\hbar m}^2 r_p(\hbar m + \hbar, t) \right) \nabla_{\hbar m} \leq - \sum_{p=1}^n \sum_{j=1}^n \frac{d_{pj}}{b_j^2} \int_{\Omega} r_p^2 \nabla_{\hbar m}. \quad (8)$$

Nonlinear coupling. By the Lipschitz condition $|f_k(r_k)| \leq \ell_k |r_k|$ and Young's inequality,

$$\begin{aligned} \int_{\Omega} \sum_{k=1}^n \beta_{pk} f_k(r_k) r_p \nabla_{\hbar m} &\leq \int_{\Omega} \sum_{k=1}^n |\beta_{pk}| \ell_k |r_k| |r_p| \nabla_{\hbar m} \\ &\leq \int_{\Omega} \sum_{k=1}^n \frac{1}{2} (|\beta_{pk}| \ell_k + |\beta_{kp}| \ell_p) r_p^2 \nabla_{\hbar m}. \end{aligned} \quad (9)$$

Combining (7)–(9) and using $V = \sum_{p=1}^n \int_{\Omega} \frac{1}{2} r_p^2 \nabla_{\hbar m}$, we obtain

$${}_0^C \nabla_{\hbar}^{\theta} V \leq -\frac{1}{2} \sum_{p=1}^n \int_{\Omega} \left(2\alpha_p + \sum_{j=1}^n \frac{2d_{pj}}{b_j^2} - \sum_{k=1}^n (|\beta_{pk}| \ell_k + |\beta_{kp}| \ell_p) \right) r_p^2 \nabla_{\hbar m} = -\varsigma V.$$

By the fractional comparison principle (Lemma 3),

$$V(r(\hbar m, t), t) \leq V(r(\hbar m, 0), 0) E_{\theta, 1}^{\hbar}(-\varsigma, 0).$$

As $\lim_{t \rightarrow \infty} V(e(\hbar m, t), t) = 0$, then $\lim_{t \rightarrow \infty} r(\hbar m, t) = 0$, which means that drive system (5) is global uniform stable. Hence completing the proof.

Remark 5. The Lyapunov functional

$$V(r(\hbar m, t), t) = \sum_{p=1}^n \int_{\Omega} \frac{1}{2} r_p^2(\hbar m, t) \nabla_{\hbar m}$$

is selected because it effectively captures both the fractional-order temporal memory and the spatial diffusion effects inherent in system (5). For fractional-order dynamics, the quadratic energy form satisfies the inequality ${}_0^C \nabla_{\hbar}^{\theta} (\frac{1}{2} r^2) \leq r {}_0^C \nabla_{\hbar}^{\theta} r$, which ensures the monotonic decrease of V . Meanwhile, the spatial diffusion operator naturally leads to a dissipative term $-\int (\nabla_{\hbar m} r_p)^2 \nabla_{\hbar m}$, which can be bounded below by $-\sum_{j=1}^n \frac{1}{b_j^2} \int r_p^2 \nabla_{\hbar m}$ through the discrete Poincaré inequality. This structure allows the combined influence of damping, diffusion, and nonlinear coupling to be expressed in a unified quadratic form, facilitating the derivation of an explicit decay rate ς . Hence, the proposed Lyapunov functional is physically meaningful, mathematically tractable, and well suited for establishing the global uniform (asymptotic) stability of the fractional-order diffusion-coupled system.

Remark 6. If system (5) satisfies the criteria outlined in Theorem 1 and has an equilibrium solution of 0, the system is globally asymptotically stable and all solutions converge to an equilibrium point.

Remark 7. While zero-solution stability indicates system stability under certain conditions, practical networks are rarely in a zero state, with neurons or subsystems often exhibiting varied dynamics. Stability and synchronization differ: stability addresses a specific state, while synchronization concerns maintaining similar trajectories across systems. Due to network complexity, nonlinearity, delays, and disturbances, controller design is crucial for ensuring synchronization, especially in fields like communication and signal processing. Even with zero-solution stability, controllers are needed to improve synchronization and system adaptability.

In order to get synchronization criterion, considering system (5) as the driving model, the behavior of the response model is dominated by

$$\begin{cases} {}^C_0\nabla_h^\theta u_p(\hbar m, t) &= \sum_{j=1}^n d_{pj} \nabla_{\hbar m}^2 u(\hbar m + \hbar, t) - \alpha_p u_p(\hbar m, t) \\ &+ \sum_{k=1}^n \beta_{pk} f_k(u_k(\hbar m, t)) + h_p(\hbar m, t) + J_p, \\ \nabla_{\hbar m} u_p(\hbar m + \hbar, t) |_{\hbar m \in \partial\Omega} &= 0, u_p(\hbar m, 0) = \zeta_p(\hbar m), (\hbar m, t) \in (\Omega \times (h\mathbb{N})_h), \end{cases} \quad (10)$$

where $h_p(\hbar m, t)$ is controller, $\nabla_{\hbar m} u_p(\hbar m + \hbar) |_{\hbar m \in \partial\Omega} = 0$ is boundary condition of system (10).

The error $e_p(\hbar m, t) = u(\hbar m, t) - v(\hbar m, t)$ between the driving system and the response system is characterized as

$$\begin{cases} {}^{CT}_0\nabla_h^{\theta, \varpi} e_p(\hbar m, t) &= \sum_{j=1}^n d_{pj} \nabla_{\hbar m}^2 e(\hbar m + \hbar, t) - \alpha_p e_p(\hbar m, t) \\ &+ \sum_{k=1}^n \beta_{pk} f_k(e_k(\hbar m, t)) + h_p(\hbar m, t), \\ \nabla_{\hbar m} e_p(\hbar m + \hbar, t) |_{\hbar m \in \partial\Omega} &= 0, e_p(\hbar m, 0) = \phi_p(\hbar m), (\hbar m, t) \in (\Omega \times (h\mathbb{N})_h), \end{cases} \quad (11)$$

where $\nabla_{\hbar m} e_p(\hbar m + \hbar) |_{\hbar m \in \partial\Omega} = 0$ is boundary condition of system (11). Choose the feedback control strategy as follows

$$h_p(\hbar m, t) = -\partial_p e_p(\hbar m, t). \quad (12)$$

Definition 5. system (5) and (10) are termed as globally M-L synchronized if there exist $\eta > 0$, and $0 < \xi < 1$ such that

$$\|u_p(\hbar m, t) - v_p(\hbar m, t)\| \leq \|\varphi_p(\hbar m) - \zeta_p(\hbar m)\| E_{\theta, 1}^h(-\xi, 0).$$

In summary, the model captures the essential dynamics of the system, including spatial diffusion and external influences. However, to ensure desired collective behavior across networks, synchronization control becomes necessary. The following section focuses on the design and analysis of such control strategies

Theorem 2. Under Assumption 1 and controller (12), the following condition holds

$$0 < \xi = \min_{1 \leq p \leq n} \left\{ 2\alpha_p + 2\partial_p + \sum_{j=1}^n \frac{2d_{pj}}{b_j^2} - \sum_{k=1}^n \{ |\beta_{pk}| \mathcal{L}_k + |\beta_{kp}| \mathcal{L}_p \} \right\} \leq h^{-\theta}.$$

Then, systems (5) and (10) are globally Mittag-Leffler synchronized.

Proof. Constructing the Lyapunov function as follows

$$V(e(\hbar m, t), t) = \sum_{p=1}^n \int_{\Omega} \frac{1}{2} (e_p(\hbar m, t))^2 \nabla_{\hbar m}.$$

In the light of Lemma 1 and (11)-(12), we have

$$\begin{aligned}
 {}^C_0\nabla_h^\theta V(e(\hbar m), t) &= {}^C_0\nabla_h^\theta \sum_{p=1}^n \int_{\Omega} \frac{1}{2} (e_p(\hbar m, t))^2 \nabla_{\hbar m} \\
 &\leq \sum_{p=1}^n \int_{\Omega} e_p(\hbar m, t) {}^C_0\nabla_h^{\theta, \mu} e_p(\hbar m, t) \nabla_{\hbar m} \\
 &= \sum_{p=1}^n \int_{\Omega} e_p(\hbar m, t) \left\{ \sum_{j=1}^n d_{pj} \nabla_{\hbar m}^2 e(\hbar m + \hbar, t) - \alpha_p e_p(\hbar m, t) \right. \\
 &\quad \left. + \sum_{k=1}^n \beta_{pk} f_k(e_k(\hbar m, t)) - \vartheta_p e_p(\hbar m, t) \right\} \nabla_{\hbar m} \\
 &= \sum_{p=1}^n \int_{\Omega} e_p(\hbar m, t) \sum_{j=1}^n d_{pj} \nabla_{\hbar m}^2 e(\hbar m + \hbar, t) \nabla_{\hbar m} \\
 &\quad - \sum_{p=1}^n \int_{\Omega} \alpha_p e_p^2(\hbar m, t) \nabla_{\hbar m} - \sum_{p=1}^n \int_{\Omega} \vartheta_p e_p^2(\hbar m, t) \nabla_{\hbar m} \\
 &\quad + \sum_{p=1}^n \int_{\Omega} \sum_{k=1}^n \beta_{pk} f_k(e_k(\hbar m, t)) e_p(\hbar m, t) \nabla_{\hbar m}.
 \end{aligned} \tag{13}$$

By using the distribution summation formula and boundary condition Lemma 2, we obtain

$$\begin{aligned}
 &\int_{\Omega} e_p(\hbar m, t) \sum_{j=1}^n d_{pj} \nabla_{\hbar m}^2 e(\hbar m + \hbar, t) \nabla_{\hbar m} \\
 &= \sum_{j=1}^n d_{pj} r(\hbar m + \hbar, t) \nabla_{\hbar m} e(\hbar m + \hbar, t) \Big|_{-b_j}^{b_j} - \int_{\Omega} \sum_{j=1}^n d_{pj} (\nabla_{\hbar m} e(\hbar m + \hbar, t))^2 \nabla_{\hbar m} \\
 &= - \sum_{\hbar m = -b_j + \hbar}^{b_j} \sum_{j=1}^n d_{pj} (\nabla_{\hbar m} e(\hbar m + \hbar, t))^2 \nabla_{\hbar m} \\
 &= - \sum_{\hbar m = -b_j + \hbar}^{b_j} \sum_{j=1}^n d_{pj} (\nabla_{\hbar m} e(\hbar m, t))^2 \nabla_{\hbar m} + \sum_{j=1}^n d_{pj} (\nabla_{-b_j + \hbar} e(-b_j + \hbar, t))^2 \nabla_{\hbar m} \\
 &\quad - \sum_{j=1}^n d_{pj} (\nabla_{b_j + \hbar} e(b_j + \hbar, t))^2 \nabla_{\hbar m} \\
 &= - \int_{\Omega} \sum_{j=1}^n d_{pj} (\nabla_{\hbar m} e(\hbar m, t))^2 \nabla_{\hbar m} \\
 &\leq - \sum_{j=1}^n \frac{d_{pj}}{b_j^2} \int_{\Omega} (e(\hbar m, t))^2 \nabla_{\hbar m}.
 \end{aligned}$$

Therefore, we get

$$\int_{\Omega} e_p(\hbar m, t) \sum_{j=1}^n d_{pj} \nabla_{\hbar m}^2 e(\hbar m + \hbar, t) \nabla_{\hbar m} \leq - \sum_{j=1}^n \frac{d_{pj}}{b_j^2} \int_{\Omega} (e(\hbar m, t))^2 \nabla_{\hbar m}. \tag{14}$$

According to Assumption 1, we have

$$\begin{aligned}
 &\sum_{p=1}^n \int_{\Omega} \sum_{k=1}^n \beta_{pk} f_k(e_k(\hbar m, t)) e_p(\hbar m, t) \nabla_{\hbar m} \\
 &\leq \sum_{p=1}^n \int_{\Omega} \sum_{k=1}^n |\beta_{pk}| \mathcal{L}_k |e_k(\hbar m, t)| \|e_p(\hbar m, t)\| \sum_{p=1}^n \nabla_{\hbar m} \\
 &\leq \frac{1}{2} \sum_{p=1}^n \int_{\Omega} \sum_{k=1}^n |\beta_{pk}| \mathcal{L}_k (e_k(\hbar m, t))^2 \nabla_{\hbar m} \\
 &\quad + \frac{1}{2} \sum_{p=1}^n \int_{\Omega} \sum_{k=1}^n |\beta_{pk}| \mathcal{L}_k (e_p(\hbar m, t))^2 \nabla_{\hbar m} \\
 &= \sum_{p=1}^n \int_{\Omega} \sum_{k=1}^n \frac{1}{2} \{ |\beta_{pk}| \mathcal{L}_k + |\beta_{kp}| \mathcal{L}_p \} (e_p(\hbar m, t))^2 \nabla_{\hbar m}.
 \end{aligned} \tag{15}$$

From (13) to (15), we obtain

$$\begin{aligned} {}^C_0\nabla_h^\theta V((e(hm), t), t) &\leq -\frac{1}{2} \sum_{p=1}^n \int_{\Omega} (2\alpha_p + 2\vartheta_p + \sum_{j=1}^n \frac{d_{pj}}{b_j^2} \\ &\quad - \sum_{k=1}^n \{ |\beta_{pk}| \ell_k + |\beta_{kp}| \ell_p \}) (e_p(hm, t))^2 \nabla_{hm} \\ &= -\xi \sum_{p=1}^n \int_{\Omega} \frac{1}{2} (e_p(hm, t))^2 \nabla_{hm} \\ &= -\xi V(e(hm, t), t), \end{aligned}$$

where $\xi = \min_{1 \leq p \leq n} \{ 2\alpha_p + 2\vartheta_p + \sum_{j=1}^n \frac{d_{pj}}{b_j^2} - \sum_{k=1}^n \{ |\beta_{pk}| \ell_k + |\beta_{kp}| \ell_p \} \} \geq 0$. Therefore, based on Lemma 3

$$V(e(hm, t), t) \leq V(e(hm, 0), 0) E_{\theta, 1}^h(-\xi, 0).$$

As $\lim_{t \rightarrow \infty} V(e(hm, t), t) = 0$, then $\lim_{t \rightarrow \infty} e(hm, t) = 0$, which means that drive system (5) and response system (10) achieve global asymptotical synchronization under the controller (12). Hence completing the proof.

Remark 8. The finite central difference method is widely recognized as an important approach for solving partial differential equations in numerical computation. In recent years, numerous studies have focused on spatial discrete models across various fields of science and engineering, often referred to as lattice models (see papers [37,38]). While several reports have explored neural networks with reaction–diffusion [14,28,33–35], there is, to the authors' knowledge, a lack of research on discrete-time fractional-order spatial and temporal neural networks. This paper aims to fill this gap in the existing literature.

Remark 9. The linear state feedback control law adopted in Theorem 2 is designed to establish a clear, verifiable, and analytically tractable synchronization framework for fractional-order diffusion networks. Although the controller $h_p(hm, t) = -\vartheta_p e_p(hm, t)$ appears simple, its structure is deliberately formulated to ensure that the stability analysis remains manageable under the Caputo–nabla fractional operator and spatial diffusion coupling. The control gain ϑ_p directly affects the decay rate ξ in the synchronization inequality; increasing ϑ_p strengthens the dissipative effect in the Lyapunov derivative, thereby accelerating the synchronization process. Thus, ϑ_p acts as an essential tuning parameter for regulating synchronization performance by quantifying the convergence rate of the synchronization error.

Remark 10. The synchronization condition in Theorem 2 incorporates the control gain ϑ_p , diffusion coefficients d_{pj} , damping factors α_p , coupling terms β_{pk} , and Lipschitz constants ℓ_k within a unified Lyapunov framework, providing clear analytical interpretability. The criterion states that synchronization is guaranteed when the total dissipative term $(2\alpha_p + 2\vartheta_p + \sum_j 2d_{pj}/b_j^2)$ dominates the coupling-induced amplification term $\sum_k (|\beta_{pk}| \ell_k + |\beta_{kp}| \ell_p)$. Accordingly, the criterion enables direct assessment of how variations in diffusion strength, reaction rates, or the fractional order θ influence the stability margin. Although the current result assumes known system parameters and a fixed network topology to highlight core fractional and spatial diffusion mechanisms, the framework can be extended to uncertain or time-varying networks by introducing adaptive or robust gain adjustment strategies. Therefore, the synchronization criterion serves not only as a theoretical guarantee but also as a scalable foundation for future studies involving more complex, high-dimensional, and perturbed networked systems.

Remark 11. Compared to Refs. [34,35], this paper studies the synchronization control of discrete-time fractional-order reaction–diffusion neural networks. The inclusion of fractional order makes our model more general. However, unlike the integer-order treatment in Refs. [34,35], we use Lemma 3 and the properties of the Mittag-Leffler function to derive synchronization conditions. Furthermore, our approach to the diffusion term differs from the treatments in [34,35]. By proving Lemma 2, we provide a new method for handling the diffusion term, laying the foundation for subsequent synchronization control of discrete-time fractional-order neural networks.

4. Numerical simulation

To address reproducibility and ensure clarity in the numerical implementation, this subsection provides the complete computational configuration adopted in Examples 1 and 2. All simulations were carried out using MATLAB R2023a with a fixed-step Grünwald–Letnikov numerical scheme to approximate the Caputo fractional difference operator. The total simulation time was set to $t \in [0, 100]$ with a constant integration step size of $\Delta t = 0.001$.

The considered neural network consists of two coupled nodes ($n = 2$) distributed over a one-dimensional discretized spatial domain. The spatial settings are defined as:

Example 1: $m \in [-1, 1]$, $\Delta m = 0.01$,

Example 2: $m \in [-2, 2]$, $\Delta m = 0.05$.

A zero-flux Neumann boundary condition was imposed to prevent artificial energy leakage at the boundaries. Formally, the imposed boundary condition is given by

$$\nabla_m v_p(m, t) \Big|_{m=m_{\min}} = 0, \quad \nabla_m v_p(m, t) \Big|_{m=m_{\max}} = 0, \quad p = 1, 2,$$

Table 1

Fractional operator and numerical settings used in the simulations.

Example	Fractional order	Spatial step h	Numerical scheme
Example 1	$\gamma = 0.98$	$h = 0.08$	Grünwald–Letnikov
Example 2	$\gamma = 0.90$	$h = 0.05$	Grünwald–Letnikov

and numerically implemented via a mirror-reflection mechanism:

$$v_p(m_{\min} - \Delta m, t) = v_p(m_{\min}, t), \quad v_p(m_{\max} + \Delta m, t) = v_p(m_{\max}, t).$$

The fractional-order operators used in the simulations are summarized in [Table 1](#).

To avoid layout issues caused by floating placement, the configuration parameters are listed below without floating behavior.

The diffusion (coupling) matrix used in both numerical examples is defined as

$$D = \begin{bmatrix} 0.75 & 0 \\ 0 & 0.75 \end{bmatrix},$$

and the nonlinear neuron activation function is chosen as $\tanh(\cdot)$.

For [Example 2](#), synchronization between the drive system and the response system was achieved using the control input

$$h_p(m, t) = -\partial_p e_p(m, t), \quad p = 1, 2,$$

where $\partial_1 = 0.7$, $\partial_2 = 0.6$, and the Lipschitz constants were set to $L_1 = L_2 = 1$.

The above setup provides all necessary implementation details to ensure the numerical results in [Figs. 1–8](#) can be reliably reproduced under identical computational conditions.

Example 1. The networks are modeled as follows:

$$\begin{aligned} {}^C \nabla_{0.08}^{0.98} v_p(0.01 m, t) &= \sum_{j=1}^n d_{pj} \nabla_{0.08}^{0.98} v(0.01 m + 0.08, t) - \alpha_p v_p(0.01 m, t) \\ &+ \sum_{k=1}^n \beta_{pk} \tanh(v_k(0.01 m, t)) + J_p, \end{aligned} \quad (16)$$

The parameters of system (16) are $p = 1, 2$, $d_{11} = 0.75$, $d_{12} = d_{21} = 0$, $d_{22} = 0.75$, $\alpha_1 = 1$, $\alpha_2 = 0.6$, $\beta_{11} = 0.28$, $\beta_{12} = -0.51$, $\beta_{21} = 1.31$, $\beta_{22} = -1.23$, $J_p = 0$.

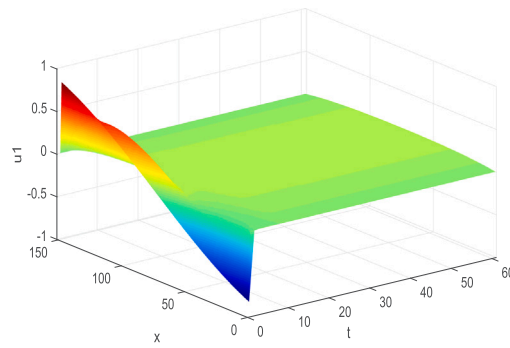
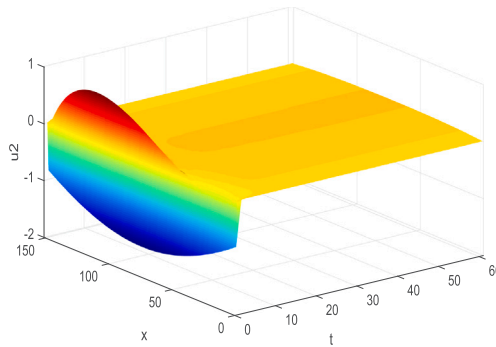
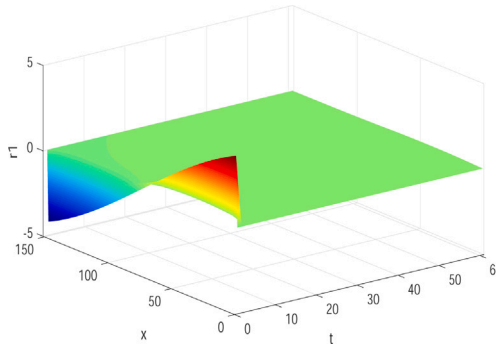
After performing the calculations, we obtain $0 < \varsigma = 1.1 < 0.08^{-0.98} \approx 11.884$, which satisfies the conditions outlined in [Theorem 1](#). The initial conditions for system (16) are selected as $v_1(0.01 m, t) = 6 \sin(-1 + t + 0.01 m)$, $v_2(0.01 m, t) = 1.5 \sin(-1 + t + 0.01 m)$, for $-1 \leq 0.01 m \leq 1$. [Figs. 1 and 2](#) show the convergent behaviors of system (16) with the initial conditions $v_1(0.01 m, t)$ and $v_2(0.01 m, t)$, respectively. These figures illustrate how the system's state variables evolve over time, demonstrating the stability and convergence properties of the proposed model. By depicting the decay of transient responses, they provide insight into the effectiveness of the designed control strategy in ensuring asymptotic stability. Furthermore, [Figs. 1 and 2](#) help visualize the influence of different initial conditions on the system's dynamics, reinforcing the theoretical analysis and supporting the validity of the derived stability criteria.

The alternative initial conditions for system (16) are given as $v_1(0.01 m, t) = \sin(-1 + t + 0.01 m)$, $v_2(0.01 m, t) = -1.5 \cos(-1 + t + 0.01 m)$, for $-1 \leq 0.01 m \leq 1$. [Figs. 3 and 4](#) demonstrate the convergent behaviors of system (16) with these initial conditions $r_1(0.01 m, t)$ and $r_2(0.01 m, t)$. The trajectory errors of system (16) for the two different sets of initial conditions are illustrated in [Figs. 3–4](#), confirming that system (16) is globally uniformly stable. These figures provide a clear comparison of the system's error convergence behavior, demonstrating its robustness against different initial states. By visualizing the error dynamics, [Figs. 3–4](#) highlight the effectiveness of the proposed stability criteria and validate the theoretical findings, offering valuable insights into the system's long-term behavior under varying conditions.

Example 2. The system is modeled as follows

$$\begin{aligned} {}^C \nabla_{0.05}^{0.9} v_p(0.05 m, t) &= \sum_{j=1}^n d_{pj} \nabla_{0.05}^{0.9} v(0.05 m + 0.05, t) - \alpha_p v_p(0.05 m, t) \\ &+ \sum_{k=1}^n \beta_{pk} \tanh(v_k(0.05 m, t)) + J_p, \end{aligned} \quad (17)$$

The parameters of system (17) are $p = 1, 2$, $d_{11} = 0.75$, $d_{12} = d_{21} = 0$, $d_{22} = 0.75$, $\alpha_1 = 0.35$, $\alpha_2 = 0.23$, $\beta_{11} = 0.55$, $\beta_{12} = -0.02$, $\beta_{21} = 0.13$, $\beta_{22} = -0.01$, $J_1 = J_2 = 0.1$. The initial values of system (17) are $v_1(0.05 m, t) = 8 \sin(-1 + t + 0.05 m)$, $v_2(0.05 m, t) = 6 \sin(-1 + t + 0.05 m)$, for $-2 \leq 0.05 m \leq 2$. [Figs. 5 and 6](#) show that system (17) exhibits a chaotic attractor, demonstrating its complex and unpredictable dynamic behavior. These figures provide crucial insights into the system's nonlinear characteristics, highlighting sensitive dependence on initial conditions and the intricate structure of the attractor. By visualizing the chaotic trajectories, [Figs. 5 and 6](#) help reveal the underlying mechanisms driving the system's instability and long-term unpredictability.

Fig. 1. The convergent behaviors of $u_1(t, hm)$.Fig. 2. The convergent behaviors of $u_2(t, hm)$.Fig. 3. The convergent behaviors of $r_1(t, hm)$.

The response system of (17) is described as

$$\begin{aligned} {}^C_0\nabla_{0.05}^{0.9} u_p(0.05 m, t) &= \sum_{j=1}^n d_{pj} \nabla_{0.05m}^{0.9} u_p(0.05m + 0.05, t) - \alpha_p u_p(0.05 m, t) \\ &+ \sum_{k=1}^n \beta_{pk} \tanh(u_k(0.05 m, t)) + h_p(0.05 m, t) + J_p, \end{aligned} \quad (18)$$

where d_{pj} , α_p , β_{pk} , are identical with network (17), and $h_p(hm, t)$ is designed as (12). The initial values of system (18) are $u_1(0.05 m, t) = -5 \cos(-1 + t + 0.05 m)$, $u_2(0.05 m, t) = -6 \cos(-1 + t + 0.05 m)$, for $-2 \leq 0.05m \leq 2$.

Using the controller $h_p(hm, t) = -\vartheta_p e_p(hm, t)$ with parameters $\vartheta_1 = 0.7$ and $\vartheta_2 = 0.6$ the Lipschitz constants are set to $L_p = 1$, for $p = 1, 2$. The conditions outlined in Theorem 2 are satisfied with $0 < \xi = 0.26 < 0.5^{-0.9} \approx 1.866$. As a result, the drive-response systems achieve global asymptotic synchronization, as demonstrated in Figs. 7 and 8. These figures provide a visual representation of the synchronization process, illustrating the convergence of the response system to the drive system over time. Specifically, Figs.

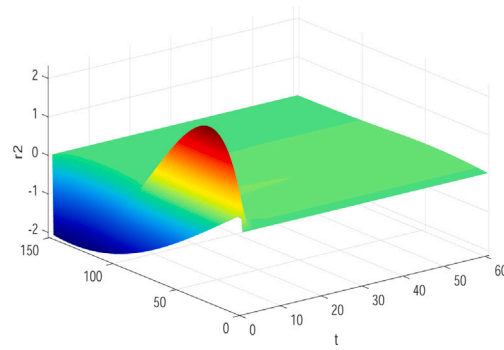


Fig. 4. The convergent behaviors of $r_2(t, hm)$.

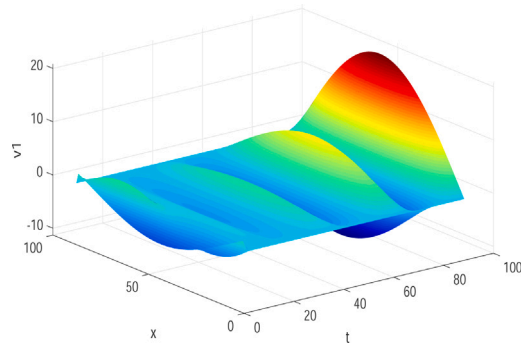


Fig. 5. The convergent behaviors of $v_1(t, hm)$.

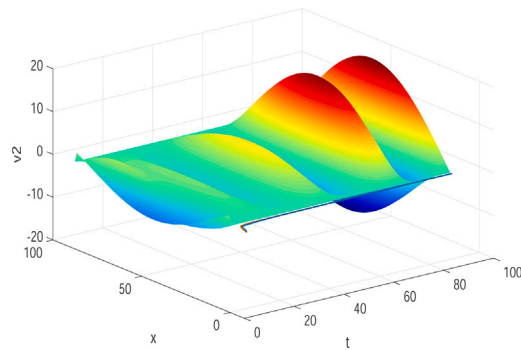


Fig. 6. The convergent behaviors of $v_2(t, hm)$.

7–8 depicts the time-domain evolution of the state variables, highlighting the diminishing synchronization error. Together, these figures validate the effectiveness of the proposed control strategy in achieving stable synchronization under the given conditions.

5. Conclusion

In conclusion, this study presents a novel approach by combining discrete diffusion terms with fractional-order neural networks, filling a significant gap in current research. Using the difference operator, we derived an integral inequality to handle diffusion terms, establishing global asymptotic stability under Lyapunov theory. The synchronization criteria with linear state feedback controllers were validated through numerical simulations, confirming the robustness of the theoretical findings. Our model has broad potential across fields. In neural dynamics, it improves neural architectures for better memory modeling and dynamic simulations. In image processing, it addresses tasks like denoising, restoration, and feature extraction, especially for noisy data. Additionally, the model shows promise in complex system applications like epidemic spread, traffic flow, and financial markets, where memory effects and non-local interactions are key. The stability criteria are also relevant for real-time control and distributed systems, ensuring stability

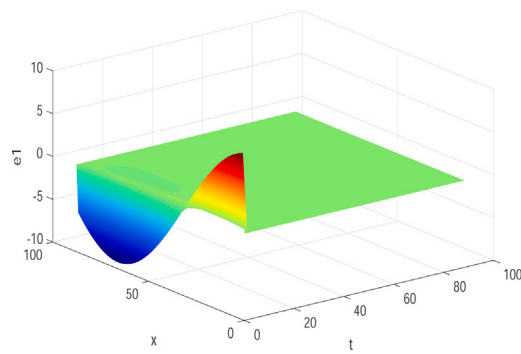


Fig. 7. The convergent behaviors of $e_1(t, hm)$.

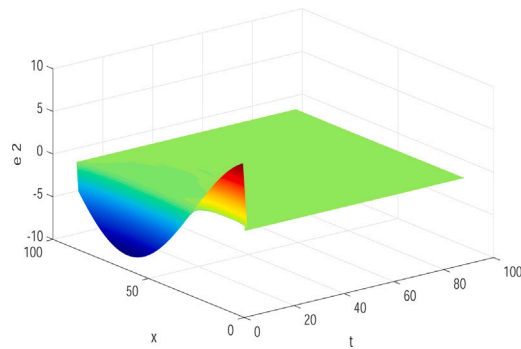


Fig. 8. The convergent behaviors of $e_2(t, hm)$.

under uncertainties. This approach offers a strong foundation for future research and practical applications in both academic and industrial settings. Future work could extend the method to other neural networks and complex systems for broader applicability.

CRediT authorship contribution statement

Xiao-Li Zhang: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Yongguang Yu:** Supervision, Resources, Methodology. **Hu Wang:** Investigation, Conceptualization. **Guojian Ren:** Visualization, Validation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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